# Sushila Yerukula

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## Professional Summary

Aspiring Data Scientist with experience in predictive modeling, healthcare analytics, and financial fraud detection. Skilled in building end-to-end machine learning pipelines including preprocessing, feature engineering, cross-validation, and performance optimization. Strong foundation in statistical analysis and model evaluation techniques.

## Education

**VIT Andhra Pradesh**, B.Tech in Computer Science and Engineering (CGPA: 8.5/10) 2022 – 2026
**Montessori School, Kurnool**, SSC (98%) 2019 – 2020

## Technical Skills

- **Programming:** Python, SQL
- **Data Analysis:** Pandas, NumPy, Exploratory Data Analysis, Feature Engineering
- **Machine Learning:** Logistic Regression, Random Forest, SVM, Cross-Validation
- **Evaluation Metrics:** Precision, Recall, F1-score, ROC-AUC
- **Deep Learning:** CNN, RNN, TensorFlow, Keras
- **Visualization:** Matplotlib, Seaborn
- **Tools:** Jupyter Notebook, Git, GitHub

## Data Science Projects

**CKD Prediction System**
GitHub
Analyzed over 400 clinical records to develop a predictive model for early Chronic Kidney Disease detection. Performed preprocessing including missing value imputation, feature scaling, and correlation analysis. Implemented Logistic Regression and Random Forest models using 5-fold cross-validation, achieving 88–90% accuracy with improved F1-score. Evaluated model performance using ROC-AUC and confusion matrix metrics. here we have done hybrid model using lstm and gru which fines future prediction

**Credit Card Fraud Detection**
GitHub
Developed machine learning models on a highly imbalanced transaction dataset for fraud detection. Applied class weighting techniques to improve recall for fraudulent cases and evaluated model robustness using ROC-AUC and Precision-Recall curve analysis.

**Food Quality Detection (CNN)**
GitHub
Designed and trained a Convolutional Neural Network for food freshness classification using over 2,000 labeled images. Applied augmentation and dropout regularization to reduce overfitting, achieving approximately 90% validation accuracy.

## Research Publication

**Comparative Analysis of Machine Learning and Deep Learning Models for Credit Card Fraud Detection**
*IEEE Conference Publication, 2025*
View Publication
Conducted a comparative evaluation of Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, k-Nearest Neighbors, and Recurrent Neural Networks for credit card fraud detection. Assessed model performance using Accuracy, F1-score, and AUC-ROC metrics, identifying Random Forest and SVM as the most robust and accurate approaches for real-time financial fraud detection systems.

## Certifications

- Oracle Cloud Infrastructure 2025 – DevOps Professional (Oracle)
- Oracle Cloud Infrastructure 2025 – Foundations Associate (Oracle)
- Adobe UI/UX Design Certification
- Android Development – E-Cell IIT Roorkee

## Coding Profiles

LeetCode: leetcode.com/u/Jayamma — HackerRank: hackerrank.com/profile/susheelay160